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TURNTABLE DRIVING DEVICE

Abstract

A turntable driving device is electrically connected to an audio player and includes a base seat disposed on the audio player, a hollowed cylinder connected fixedly to the base seat, a rotating disc module including a driving unit and a rotatable plate, an axial bearing disposed between the base seat and the driving unit, and a jog wheel module rotatably sleeved on a top portion of the cylinder. The driving unit is sleeved rotatably on a bottom portion of the cylinder, and the rotatable plate is mounted to and driven by the driving unit to rotate. The jog wheel module is operable to change between a co-movable state, where the jog wheel module is co-rotatable with the rotatable plate, and a non-co-movable state, where the jog wheel module is rotatable relative to the rotatable plate.

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Background/Summary

FIELD

[0001] The disclosure relates to a driving device, and more particularly to a turntable driving device for an audio player.

BACKGROUND

[0002] Generally, in order to provide a particular atmosphere suitable for a particular occasion, for example, to create a lively atmosphere at a party, a disc jockey (DJ) moves by hand a vinyl record back and forth to control rotation of the vinyl record, thereby adjusting a playback speed of music or repeating a certain phrase of music to generate a special sound effect. Moreover, volume of music may also be adjusted to facilitate creation of a lively atmosphere.

[0003] In nowadays, music is usually stored in a digital form in an audio file that is stored in a machine readable medium, and an audio player capable of accessing and playing the audio file stored in the machine readable medium is connected thereto to play music. For example, the audio player is electrically connected to a mobile phone or a hard disk to read the audio file stored therein to thereby play music.

SUMMARY

[0004] Therefore, an object of the disclosure is to provide a turntable driving device that can emulate a feedback similar to scratching a vinyl record.

[0005] According to the disclosure, a turntable driving device is provided. The turntable driving device is adapted to be electrically connected to an audio player that is accessible to a machine readable storage medium storing a music information therein and that is for playing the music information. The turntable driving device includes a base seat, a cylinder, a rotating disc module, an axial bearing, and a jog wheel module. The base seat is adapted to be disposed on the audio player. The cylinder is hollowed and is connected fixedly to the base seat. The rotating disc module includes a driving unit sleeved rotatably on a bottom portion of the cylinder, and a rotatable plate mounted to and driven by the driving unit to rotate. The axial bearing is disposed between the base seat and the driving unit. The jog wheel module is rotatably sleeved on a top portion of the cylinder, and is operable to change between a co-movable state, where the jog wheel module is co-rotatable with the rotatable plate, and a non-co-movable state, where the jog wheel module is rotatable relative to the rotatable plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

[0007] FIG. **1** is a perspective view of a turntable driving device of an embodiment according to the present disclosure adapted to be mounted to an audio player.

[0008] FIG. **2** is a perspective view of the turntable driving device of the embodiment.

[0009] FIG. **3** is a partly exploded perspective view of the embodiment.

[0010] FIG. **4** is a partly exploded perspective view of the embodiment from another view of angle different from FIG. **3**.

[0011] FIG. **5** is a top view of the embodiment.

[0012] FIG. **6** is a sectional view taken along line VI-VI in FIG. **5**.

[0013] FIG. **7** is a cross-sectional view of the embodiment, illustrating a rotatable plate sensing module of the turntable driving device.

[0014] FIG. **8** is a sectional view taken along line VIII-VIII in FIG. **5**.

[0015] FIG. **9** is a cross-sectional view of the embodiment, illustrating a jog wheel sensing module of the turntable driving device.

[0016] FIG. **10** is a sectional view taken along line X-X in FIG. **5**.

DETAILED DESCRIPTION

[0017] Before the disclosure is described in greater detail, it should be noted that where considered appropriate, reference numerals or terminal portions of reference numerals have been repeated among the figures to indicate corresponding or analogous elements, which may optionally have similar characteristics.

[0018] It should be noted herein that for clarity of description, spatially relative terms such as "top," "bottom," "upper," "lower," "on," "above," "over," "downwardly," "upwardly" and the like may be used throughout the disclosure while making reference to the features as illustrated in the drawings. The features may be oriented differently e.g., rotated 90 degrees or at other orientations and the spatially relative terms used herein may be interpreted accordingly.

[0019] Referring to FIGS. **1** to **3**, a turntable driving device **10** of an embodiment according to the present disclosure is adapted to be electrically connected to an audio player **90**. The audio player **90** is accessible to a machine readable storage medium (not shown) that stores a music information therein and is for playing the music information. It should be noted that the music information may be stored in another external storage device (not shown) that is signally connected to the audio player **90**, and the external storage device is, for example, a handheld electronic device or a flash drive.

[0020] Referring to FIGS. **2** to **4**, the turntable driving device **10** includes a base seat **1**, a cylinder **2**, a rotating disc module **3**, an axial bearing **4**, a jog wheel module **5**, a display module **6**, a rotatable plate sensing module **7** (see FIG. **7**), and a jog wheel sensing module **8** (see FIG. **9**). The cylinder **2** extends vertically in an axial direction (A) and is connected fixedly to the base seat **1**. The rotating disc module **3** is sleeved on the cylinder **2**. The axial bearing **4** is disposed between the base seat **1** and the rotating disc module **3**. The jog wheel module **5** is rotatably sleeved on a top portion of the cylinder **2**.

[0021] Referring to FIGS. **1**, **3**, **4** and **6**, the base seat **1** is adapted to be disposed on the audio player **90**, and includes a base portion **11** and a protruding portion **12** extending upwardly from the base portion **11** in the axial direction (A), and is formed with a through hole **13** extending vertically through the protruding portion **12** in the axial direction (A). The cylinder **2** is hollowed, is inserted into the through hole **13** of the base seat **1**, and is fixed to a bottom surface of the base seat **1** by a hexagonal nut **91**. In this embodiment, the base portion **11** includes a seat body **111**, and a circuit board **112** disposed on the seat body **111** and signally connected to the audio player **90** (see FIG. **1**). It should be noted that the circuit board **112** may be integrated with a printed circuit board (not shown) of the audio player **90** or may be signally connected to the audio player **90** via a transmission line according to practical application requirements.

[0022] The rotating disc module **3** includes a driving unit **31** sleeved rotatably on the cylinder **2**, a rotatable plate **32** mounted to and driven by the driving unit **31** to rotate, and a wear-resistant set **33**.

[0023] The driving unit **31** includes a first rotating shaft **311** rotatably sleeved on a bottom portion of the cylinder **2**, and a motor **312** mounted under the rotatable plate **32** and surrounding the first rotating shaft **311**. The rotatable plate **32** is mounted coaxially to and connected fixedly to the first rotating shaft **311** and is connected to and driven by the motor **312** to rotate so as to drive rotation of the first rotating shaft **311**. The first rotating shaft **311** includes a head portion **313** rotatably sleeved on and radially spaced apart from the cylinder **2**, and an extension portion **314** integrally and downwardly extending from the head portion **313** in the axial direction (A) and sleeved on the cylinder **2**. The extension portion **314** has an outer diameter smaller than an outer diameter of the head portion **313** in a radial direction perpendicular to the axial direction (A).

[0024] The motor **312** is adapted to be electrically connected to the audio player **90**, and includes a stator **315** connected fixedly to the base seat **1** and surrounding an outer peripheral surface of the first rotating shaft **311**, and a rotor sleeve **316** surrounding and spaced apart from an outer peripheral surface of the stator **315**, and rotating about the stator **315** when the stator **315** is energized by a power supply, e.g., an alternating current. In this embodiment, the stator **315** is a coil assembly, and the rotor sleeve **316** is a magnet assembly and the present disclosure is not limited in this respect.

[0025] The rotatable plate **32** has an annular top surface **321**, and a circular groove **322** indented downwardly from an inner periphery of the annular top surface **321**. The rotatable plate **32** is fixedly connected to top portions respectively of the rotor sleeve **316** and the first rotating shaft **311** by a plurality of fasteners **92**, e.g., but not limited to, bolts. In this way, the rotor sleeve **316** drives rotations of the rotatable plate **32** and the first rotating shaft **311** when being rotated.

[0026] The wear-resistant set **33** includes two wear-resistant pads **331** that are annular and that are stacked on the annular top surface **321** of the rotatable plate **32**. In this embodiment, each of the wear-resistant pads **331** is made of polyester film but not limited in this respect.

[0027] The axial bearing **4** is sleeved directly on and surrounds the extension portion **314** of the first rotating shaft **311**, and is clamped between a top portion of the protruding portion **12** of the base seat **1** and a bottom portion of the head portion **313** of the first rotating shaft **311** in the axial direction (A). The axial bearing **4** has a rotating surface **41** in contact with the head portion **313** of the first rotating shaft **311**, and a fixed surface **42** in contact with the protruding portion **12** of the base seat **1**. By virtue of the axial bearing **4** that distributes weights of components of the turntable driving device **10** in an upright direction, i.e., the axial direction (A), the rotating disc module **3** may be smoothly rotated relative to the base seat **1** so as to prevent the rotatable plate **32** from wobbling or inclining relative to the base seat **1**.

[0028] As shown in FIG. **6**, the jog wheel module **5** includes a second rotating shaft **51** rotatably sleeved on the top portion of the cylinder **2**, a lower seat **52** sleeved on an outer periphery of the second rotating shaft **51** and connected fixedly to and co-movably with the second rotating shaft **51**, and a scratchable disc unit **53** detachably and magnetically connected to a top portion of the lower seat **52**.

[0029] Specifically, the second rotating shaft **51** is spaced apart from and disposed above the first rotating shaft **311**. The lower seat **52** includes a lower seat body **521** connected fixedly to the second rotating shaft **51**, and a plurality of first magnetic members **522** disposed on the lower seat body **521**. The lower seat body **521** has a bottom seat portion **523** and a plurality of connecting posts **524** extending upwardly from an outer periphery of the bottom seat portion **523** and angularly spaced-apart from each other. The first magnetic members **522** are disposed on top portions respectively of the top connecting posts **524**, respectively.

[0030] The scratchable disc unit **53** includes a turntable body **531** disposed above the lower seat **52**, and a plurality of second magnetic members **532** disposed on the turntable body **531**. The turntable body **531** has a transparent central region **533** that is light transmissive and a surrounding disc region **534** that extends radially and outwardly from the transparent central region **533**. The second magnetic members **532** are connected fixedly to a bottom side of the surrounding disc region **534** and are detachably and magnetically connected to the first magnetic members **522**, respectively. The wear-resistant pads **331** of the wear-resistant set **33** are sandwiched between the annular top surface **321** of the rotatable plate **32** and a bottom surface of the surrounding disc region **534** so as to facilitate rotation of the scratchable disc unit **53** relative to the rotatable plate **32**.

[0031] The jog wheel module **5** is operable to change between a co-movable state and a non-co-movable state.

[0032] Specifically, the jog wheel module **5** is co-rotatable with the rotatable plate **32** that is driven to rotate by the driving unit **31** when being in the co-movable state as a result of frictional force formed therebetween. The jog wheel module **5** is rotatable relative to the rotatable plate **32** when being in the non-co-movable state. The wear-resistant set **33** is clamped between the rotatable plate **32** and the scratchable disc unit **53** and allows rotation of the scratchable disc unit **35** relative to the rotatable plate **32** when the jog wheel module **5** is in the co-movable state. When the scratchable disc unit **53** is moved in a direction the same as or opposite to a rotational direction of the rotatable plate **32** to an extent that the frictional force between the rotatable plate **32** and the scratchable disc unit **53** is overcome, the jog wheel module **5** is changed from the co-movable state to the non-co-movable state, and the scratchable disc unit **53** rotates independently relative to the rotatable plate **32**. In other words, the jog wheel module **5** does not rotate with the rotatable plate **32**, and may rotate in a speed slower than a speed of the rotatable plate **32**, stop rotating, or rotate in a direction opposite to the rotational direction of the rotatable plate **32**.

[0033] The display module **6** includes a mounting seat **60**, a display **61** connected fixedly to the top portion of the cylinder **2**, a plastic sheet **62** disposed above the display **61**, and a transmission line set **63** connected signally to the display **61**, extending from a lower side of the base seat **1** through the cylinder **2**, and adapted to be signally connected to the audio player **90**. The plastic sheet **62** has a window portion **621** corresponding in position to the transparent central region **533**. The display **61** is disposed under the transparent central region **533** and the window portion **621**. In this embodiment, the transmission line set **63** is a flexible flat cable detachably inserted into the circuit board **112** or a circuit board (not shown) of the audio player **90**, and is signally connected to the audio player **90** and the display **61** to transceive display signals to and from the display **61**.

[0034] Referring to FIGS. **1** and **6** to **8**, the rotatable plate sensing module **7** is disposed in the base portion **11**, and includes a first light shielding sheet **71** and a first light sensing element **72**. The first light shielding sheet **71** is connected co-rotatably to the first rotating shaft **311**, is configured as a transparent circular sheet, and has a plurality of angularly spaced-apart first graduations **711** marked thereon. The first light sensing element **72** is electrically connected to the circuit board **112** so as to be signally connected to the audio player **90**, and is configured to detect a position of the first light shielding sheet **71** and output a first rotary information indicating a rotated information of the rotatable plate **32** based on the position of the first light shielding sheet **71**. When the first light shielding sheet **71** is driven by the first rotating shaft **311** to rotate, the first light sensing element **72** detects a position of the first light shielding sheet **71** based on the first graduations **711** and outputs the first rotary information, e.g., a rotational speed, of the first light shielding sheet **71** and thus a rotational speed of the first rotating shaft **311** to the circuit board **112**. In this embodiment, the first light sensing element **72** is a light blocking sensor, and may be a light reflective sensor. It should be noted that, in other embodiments, the first light sensing element **72** may be directly and electrically connected to the audio player **90**.

[0035] Referring to FIGS. **1**, **6**, **9** and **10**, the jog wheel sensing module **8** has a structure similar to that of the rotatable plate sensing module **7**, is disposed on the lower seat body **521** of the lower seat **52**, and includes a second light shielding sheet **81** and a second light sensing element **82**. The second light shielding sheet **81** is connected co-rotatably to the lower seat **521**, is configured as a transparent circular sheet, and has a plurality of angularly spaced-apart second graduations **811** marked thereon. The second light sensing element **82** is connected to the mounting seat **60** of the display module **6**, is electrically connected to the circuit board **112** so as to be signally connected to the audio player **90**, and is configured to detect a position of the second light shielding sheet **81** and output a second rotary information indicating a rotated information of the turntable body **531** based on the position of the second light shielding sheet **81** and thus a rotated information of the lower seat body **521**. When the second light shielding sheet **81** is driven by the lower seat body **521** to rotate, the second light sensing element **82** detects a position of the second light shielding sheet **81** based on the second graduations **811** and outputs the second rotary information, e.g., a rotational speed, of the second light shielding sheet **81** to the circuit board **112**. In this embodiment, the second light sensing element **82** is similar to the first light sensing element **72**, so further details of the second light sensing element **82** is omitted. It should be noted that, in other embodiments, the second light sensing element **82** may be directly and electrically connected to the audio player **90**.

[0036] To use the turntable driving device **10** of the present disclosure, the motor **312** is first energized to drive rotation of the rotatable plate **32**, by virtue of the frictional force formed between one of the wear-resistant pads **331** and the scratchable disc unit **53** and between another one of the wear-resistant pads **331** and the rotatable plate **32**, the rotatable plate **32** drives rotation of the scratchable disc unit **53**. At this time, the jog wheel module **5** is in the co-movable state. Hereafter, when a user such as a disc jockey (DJ) scratches, e.g., by hand, the scratchable disc unit **53** in a direction the same as or opposite to a rotational direction of the rotatable plate **32**, the scratchable disc unit **53** is driven to move with the hand of the DJ and is changed to the non-co-movable state. At this time, the rotational speeds and/or directions of the rotatable plate **32** and the scratchable disc unit **53** are different. By virtue of the wear-resistant pads **331** of the wear-resistant set **33**, the user may smoothly scratch the scratchable disc unit **53** to drive the rotation of the scratchable disc unit **53** in a speed slower than that of the rotatable plate **32**, so as to change the rotational direction of the scratchable disc unit **53** or stop rotation of the scratchable disc unit **53**.

[0037] Thus, the scratchable disc unit **53** may be operated by the user to emulate a feedback similar to scratching a vinyl record back and forth on a turntable.

[0038] It should be noted that, by virtue of the first light sensing element **72** of the rotatable plate sensing module **7** that detects the position of the first light shielding sheet **71** and thus rotation of the rotatable plate **32** and that outputs the first rotational information to the circuit board **112** and thus to a central processing system (not shown) of the audio player **90**, and by virtue of the second light sensing element **82** of the jog wheel sensing module **8** that detects the position of the second light shielding sheet **81** and thus rotation of the lower seat body **521** of the jog wheel module **5**, and that outputs the second rotational information to the circuit board **112** and thus to the central processing system of the audio player **90**, the central processing system of the audio player **90** processes the music information with reference to the first rotational information and the second rotational information such that the audio player **90** plays the music information processed accordingly while the motor **312** drives rotation of the rotatable plate **32** and/or the user scratches the scratchable disc unit **53**. In this way, the audio player **90** plays the music information with a special sound effect, which emulates scratching a vinyl record. It should be noted that since the main feature of the present disclosure does not reside in the processing of the music information, which is well known by person skilled in the pertinent art, and further details of the same are omitted for the sake of brevity.

[0039] In summary, by virtue of the structural design of the axial bearing **4** that is coaxially disposed between the base seat **1** and the rotating disc module **3** in the axial direction (A) and that evenly distributes weights of components of the turntable driving device **10**, a durability of the turntable driving device **10** may be improved, thus prolonging service life of the turntable driving device **10**. Moreover, since the second rotating shaft **51** and the first rotating shaft **311** are sleeved respectively on the bottom portion and the top portion of the cylinder **2** and may be detached respectively and easily therefrom, in a case where the transmission line set **63** is a flexible flat cable having a relatively large width, the cylinder **2** may be modified to enlarge its inner diameter to permit extension of the transmission line set **63** that is to be electrically connected to the audio player **90** through the cylinder **2**.

[0040] In addition, by virtue of the first magnetic members **522** and the second magnetic members **532** that are detachably and magnetically connected, the scratchable disc unit **53** may be simply detached from and connected to the lower seat **52**. In this way, it is relatively convenient to assemble and disassemble the turntable driving device **100**.

[0041] In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to “one embodiment,” “an embodiment,” an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description,

various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects; such does not mean that every one of these features needs to be practiced with the presence of all the other features. In other words, in any described embodiment, when implementation of one or more features or specific details does not affect implementation of another one or more features or specific details, said one or more features may be singled out and practiced alone without said another one or more features or specific details. It should be further noted that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.
[0042] While the disclosure has been described in connection with what is (are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

Claims

1. A turntable driving device adapted to be electrically connected to an audio player that is accessible to a machine readable storage medium storing a music information therein and that is for playing the music information, said turntable driving device comprising: a base seat that is adapted to be disposed on the audio player; a cylinder that is hollowed and that is connected fixedly to said base seat; a rotating disc module that includes a driving unit sleeved rotatably on a bottom portion of said cylinder, and a rotatable plate mounted to and driven by said driving unit to rotate; an axial bearing that is disposed between said base seat and said driving unit; and a jog wheel module that is rotatably sleeved on a top portion of said cylinder, and that is operable to change between a co-movable state, where said jog wheel module is co-rotatable with said rotatable plate, and a non-co-movable state, where said jog wheel module is rotatable relative to said rotatable plate.
2. The turntable driving device as claimed in claim 1, further comprising a display module that includes: a display connected fixedly to said top portion of said cylinder; and a transmission line set connected signally to said display, extending from a lower side of said base seat through said cylinder, and adapted to be signally connected to the audio player.
3. The turntable driving device as claimed in claim 1, wherein: said cylinder extends in an axial direction; said driving unit includes a first rotating shaft rotatably sleeved on said bottom portion of said cylinder, and a motor; and said rotatable plate is mounted coaxially to and connected fixedly to said first rotating shaft, and is connected to and driven by said motor to rotate so as to drive rotation of said first rotating shaft.
4. The turntable driving device as claimed in claim 3, wherein: said first rotating shaft includes a head portion rotatably sleeved on said cylinder, and an extension portion downwardly extending from said head portion in the axial direction, sleeved on said cylinder, and having an outer diameter smaller than an outer diameter of said head portion; and said axial bearing is sleeved on said extension portion and is clamped between said head portion and said base seat in the axial direction.
5. The turntable driving device as claimed in claim 4, wherein said motor includes: a stator connected fixedly to said base seat and surrounding said cylinder; and a rotor sleeve connected fixedly to said rotatable plate, surrounding and spaced apart from said stator, and rotating about said stator when said stator is energized.
6. The turntable driving device as claimed in claim 4, further comprising a rotatable plate sensing module that includes: a first light shielding sheet connected co-rotatably to said first rotating shaft; and a first light sensing element disposed on said base seat and configured to detect a position of

said first light shielding sheet and output a first rotary information indicating a rotated information of said rotatable plate based on the position of said first light shielding sheet.

7. The turntable driving device as claimed in claim 3, wherein: said jog wheel module includes a second rotating shaft rotatably sleeved on said top portion of said cylinder, a lower seat fixed to said second rotating shaft, and a scratchable disc unit detachably and magnetically connected to said lower seat; and said rotating disc module further includes a wear-resistant set clamped between said rotatable plate and said scratchable disc unit and allowing rotation of said scratchable disc unit relative to said rotatable plate when said jog wheel module is in the co-movable state.

8. The turntable driving device as claimed in claim 7, wherein: said lower seat includes a lower seat body connected to said second rotating shaft, and a plurality of first magnetic members disposed on said lower seat body; and said scratchable disc unit includes a turntable body disposed above said lower seat, and a plurality of second magnetic members disposed on said turntable body, and detachably and magnetically connected to said first magnetic members, respectively.

9. The turntable driving device as claimed in claim 7, further comprising: a display module; and a jog wheel sensing module that includes: a second light shielding sheet connected to said lower seat; and a second light sensing element connected to said display module and configured to detect a position of said second light shielding sheet and output a second rotary information indicating a rotated information of said scratchable disc unit based on the position of said second light shielding sheet.

10. The turntable driving device as claimed in claim 7, wherein: said scratchable disc unit includes a transparent central region that is light transmissive, and a surrounding disc region that extends radially and outwardly from said transparent central region; and said turntable driving device further comprises a display module that includes a display connected fixedly to said top portion of said cylinder and disposed under said transparent central region.